SecureLand: Project Requirements and Library Inventory

1. Document Overview

This document outlines the direct frontend packages, backend libraries, and environment prerequisites for the SecureLand project. The project utilizes npm for package management; therefore, this acts as the equivalent of a dependency manifest, rather than a Python `requirements.txt` file. The `package.json` and `package-lock.json` files remain the definitive sources of truth for installation.

2. Software and Runtime Requirements

The following core technologies are required to run, build, and serve the application:

- **Node.js (with npm):** Required for installing frontend packages and executing Vite, TypeScript, and linting scripts.
- **PHP (Version 8.0 or later):** Required to serve the backend API endpoints.
- **PHP PDO & PDO MySQL Driver (`pdo_mysql`):** Required for secure database access and query execution.
- **Database:** MySQL or MariaDB, utilized for persistent application data storage.
- **Web Server:** A server environment (e.g., Apache, Nginx, or PHP built-in server) capable of serving the PHP endpoints and the built frontend application.

Note: Database credentials must be supplied securely through the server environment variables (`SECURELAND_DB_DSN`, `SECURELAND_DB_USER`, and `SECURELAND_DB_PASSWORD`) for non-default environments. Real credentials should never be committed to the version control repository.

3. Direct Application Dependencies

The following table lists the core frontend packages required for the application's runtime interface and client-side logic.

Package	Manifest Range	Purpose
`@headlessui/react`	`^2.2.10`	Accessible, unstyled React UI components.
`lucide-react`	`^1.46.0`	Scalable icon components utilized across the interface.
`react`	`^19.2.8`	Core frontend component framework.
`react-dom`	`^19.2.8`	Renders the React component tree into the browser

4. Direct Development Dependencies

The following packages are utilized strictly during the development, styling, and build processes.

Package	Manifest Range	Purpose
`@tailwindcss/vite`	`^4.3.3`	Integrates the Tailwind CSS framework seamlessly with Vite.
`@types/node`	`^24.13.3`	TypeScript definitions for Node.js APIs used by tooling.
`@types/react`	`^19.2.18`	TypeScript definitions for React.
`@types/react-dom`	`^19.2.7`	TypeScript definitions for React DOM.
`@vitejs/plugin-react`	`^6.1.1`	React support for the Vite development server and builds.
`autoprefixer`	`^10.6.1`	Automatically appends vendor prefixes to CSS rules.
`oxlint`	`^1.81.0`	High-performance JavaScript and TypeScript linting.
`postcss`	`^8.5.28`	CSS transformation and processing tooling.
`tailwindcss`	`^4.3.3`	Utility-first CSS framework for rapid UI styling.
`typescript`	`~6.0.2`	Provides static type checking and JavaScript compilation.
`vite`	`^8.3.0`	Frontend development server and production bundler.

5. Backend Libraries and Database Driver

The backend architecture is designed to minimize external dependencies and relies heavily on native functionality:

- **No Third-Party Composer Packages:** There are no third-party PHP Composer packages declared in this project.
- **Database Connectivity:** PHP's built-in **PDO (PHP Data Objects)** API is utilized for secure, prepared-statement database access.
- **Database Driver:** PHP's **PDO MySQL** extension (`pdo_mysql`) is used to bridge PDO to the MySQL/MariaDB database.
- **Native APIs:** All other backend functions utilize PHP's native modules for session management, password verification, file uploading, filesystem I/O, JSON processing, and SHA-256 cryptographic hashing.

6. Installation and Project Commands

To initialize the project, execute the following commands from the root project directory:

Installation and Development Server:

```
npm install
```